

Def. Let  $F$  be a field. Define the General Linear Group as:

$$GL_n(F) \stackrel{\text{def}}{=} \{A \in M_{n \times n}(F) \mid \det A \neq 0\}$$

$I_n \in GL_n(F)$ ,  $\det A, \det B \neq 0 \Rightarrow \det AB \neq 0$ ,  $\det A \neq 0 \Rightarrow \det A^{-1} \neq 0$ , exists

Thus it is group.

Def. Define the Special Linear Group as:  $SL_n(F) \stackrel{\text{def}}{=} \{A \in M_{n \times n}(F) \mid \det A = 1\}$

Lemma. Let  $\mathbb{F}_q$  be a finite field of order  $q$ . Then  
 $GL_n(\mathbb{F}_q)$  has order  $(q^n - 1)(q^n - q)(q^n - q^2) \dots (q^n - q^{n-1})$

$SL_n(\mathbb{F}_q)$  has order  $\frac{1}{q-1} |GL_n(\mathbb{F}_q)|$

Proof. Clearly,  $|M_{n \times n}(\mathbb{F}_q)| = q^{n^2}$ .

Choose the first column as non-zero, there are  $q^n - 1$  possibilities.

And, the second column all but a multiple of the first, then  $q^2 - q$  possibilities.

Successively, choose the third as all but not linear combination of first and second, then there are  $q^n - q^2$  (More specifically,  $q^2$  is the number of cases of  $a_1 v_1 + a_2 v_2$ ).

Inductively, we obtain that  $|GL_n(\mathbb{F}_q)| = \prod_{k=0}^{n-1} (q^n - q^k)$

Meanwhile, observe that

$$\varphi: GL_n(\mathbb{F}_q) \rightarrow \mathbb{F}_q^\times: A \mapsto \det A$$

is well-defined (because  $\det A$  is obtained by operations on  $\mathbb{F}_q$ ), onto ( $\det \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & a \end{pmatrix} = a$ ) and Homomorphism ( $\det AB = \det A \cdot \det B$ ) with  $\ker \varphi = SL_n(\mathbb{F}_q)$ . Since

$$GL_n(\mathbb{F}_q) / \ker \varphi \cong \mathbb{F}_q^\times, \text{ we obtain } |\ker \varphi| = |SL_n(\mathbb{F}_q)| = \frac{1}{q-1} \cdot |GL_n(\mathbb{F}_q)|.$$

$$(p^2 - 1)^2 (p + 1).$$

Problem.

- 1). Find order of  $GL_2(\mathbb{F}_p)$ , where  $p$  is the prime number.
- 2). Determine  $p = \left\{ \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \mid a \in \mathbb{F}_p \right\}$  is Sylow  $p$ -subgroup of  $GL_2(\mathbb{F}_p)$ .
- 3). Find order of  $N_{GL_2(\mathbb{F}_p)}(p)$  and  $\# Syl_p(GL_2(\mathbb{F}_p))$ .

Proof.  $A \in GL_2(\mathbb{F}_p) \Leftrightarrow \det A \neq 0$ .

The possible vector in the first row in  $A$  is  $p^2 - 1$ , except zero vector.  
number of

For chosen first row, the possible number of second row is  $p^2 - p$ ,  
except scalar multiplication of the first row so,

$$|GL_2(\mathbb{F}_p)| = (p^2 - 1)(p^2 - p) = p \cdot (p-1)^2(p+1) = p(p^3 - p^2 - p + 1)$$

Clearly  $p$  has order of  $p$ , and  $p \nmid p^3 - p^2 - p + 1$ , so  $p \in Syl_p(G)$ .

Find  $N_G(p)$ . Since for any  $\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in GL_2$ , where  $ac - b \neq 0$ ,

$$\begin{pmatrix} c & -b \\ 0 & a \end{pmatrix} \cdot \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} = \frac{1}{ac}$$

$$= \begin{pmatrix} c & -b \\ 0 & a \end{pmatrix} \begin{pmatrix} a & b + ck \\ 0 & c \end{pmatrix} = \frac{1}{ac}$$

$$= \begin{pmatrix} ac & cb + c^2k - bc \\ 0 & ac \end{pmatrix} = \frac{1}{ac}$$

$$= \begin{pmatrix} 1 & cka^{-1} \\ 0 & 1 \end{pmatrix} \in p, \text{ therefore } |N_G(p)| = (p-1)^2 \cdot p.$$

$$\text{So, } n_p = [GL_2 : N_{GL_2}(p)] = p+1. \quad \square$$

Def. The Heisenberg group over  $F$  defined by

$$H(F) \stackrel{\text{def}}{=} \left\{ \begin{pmatrix} 1 & a_1 & a_2 \\ 0 & 1 & a_3 \\ 0 & 0 & 1 \end{pmatrix} \mid a_i \in F \right\} \leq SL_3(F).$$

Prop.  $Z(H(F)) \cong (F, +)$

Proof observe that

$$\underbrace{\begin{pmatrix} 1 & a_1 & a_2 \\ 0 & 1 & a_3 \\ 0 & 0 & 1 \end{pmatrix}}_A \cdot \underbrace{\begin{pmatrix} 1 & b_1 & b_2 \\ 0 & 1 & b_3 \\ 0 & 0 & 1 \end{pmatrix}}_B = \begin{pmatrix} 1 & a_1+b_1 & a_2+b_2+a_1b_3 \\ 0 & 1 & a_3+b_3 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\text{Therefore, } \begin{pmatrix} 1 & a_1 & a_2 \\ 0 & 1 & a_3 \\ 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -a_1 & -(a_2 - a_1a_3) \\ 0 & 1 & -a_3 \\ 0 & 0 & 1 \end{pmatrix}$$

And,  $AB = BA \Leftrightarrow a_1b_3 = a_3b_1$ . This implies

$$A \in Z(H(F)) \Leftrightarrow \forall b_1, b_3 \in F, a_1b_3 = a_3b_1 \\ \Leftrightarrow a_1 = a_3 = 0.$$

$$\text{Therefore, } Z(H(F)) = \left\{ \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \mid a \in F \right\}.$$

Now,  $F \rightarrow Z(H(F)): a \mapsto \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$  is isomorphism by

$$\begin{pmatrix} 1 & 0 & a \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & a+b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$GL_3(\mathbb{F}_q) \quad (q^3-1)(q^3-q)(q^3-q^2)$$

$$| \downarrow |$$

$$SL_3(\mathbb{F}_q) \quad (q^3-1)(q^3-q)q^2$$

$$| \downarrow |$$

$$H(\mathbb{F}_q) \quad q^3$$

$$| \downarrow |$$

$$Z(H(\mathbb{F}_q)) \cong (\mathbb{F}_q, +) \quad q$$

$$| \downarrow |$$

$$1.$$